A database permissions change, not an attack, stopped Cloudflare’s core traffic on 18 November 2025

“The issue was not caused, directly or indirectly, by a cyber attack or malicious activity of any kind.” — Cloudflare, 18 November 2025

11:28 → 17:06 UTC

5 h 38 min from the first customer errors to full restoration — derived from the published timeline.

Source: Cloudflare, “Cloudflare outage on November 18, 2025”, Matthew Prince. All times in this deck are UTC.

One feature file doubled in size and a fixed 200-feature limit turned it into a global 5xx event

The failure was in the ingestion of Cloudflare's own generated configuration, not in the traffic that configuration governs.

NORMAL · CHANGE DEPLOYED

11:05

Database access control change deployed

CRITICAL · IMPACT STARTS

11:28

First errors observed on customer HTTP traffic

DEGRADED · MAIN IMPACT ENDS

14:30

Correct configuration file deployed globally

RECOVERED · ALL RESOLVED

17:06

All downstream services restarted and restored

TRIGGER

A change to one of Cloudflare's database systems' permissions made the query output duplicate entries into the feature file.

MECHANISM

The file doubled in size and passed the bots module's preallocated limit of 200 features, against roughly 60 then in use.

SCOPE

Core CDN and security services, Turnstile, Workers KV, Dashboard, Email Security and Access were affected, in different ways.

RECOVERY

Core traffic was largely flowing as normal by 14:30, and all remaining services were restored at 17:06.

Cloudflare, "Cloudflare outage on November 18, 2025". All times UTC.

Six product surfaces failed, and two of them degraded rather than stopped

Severity	Service or product	Impact as published
Critical	**Core CDN and security services**	HTTP 5xx status codes returned to end users.
Critical	**Turnstile**	Turnstile failed to load.
Critical	**Workers KV**	
Critical	**Access**	
Degraded	**Dashboard**	
Degraded	**Email Security**	

Response latency also rose across the CDN during the impact window: debugging and observability systems consumed large amounts of CPU automatically enriching uncaught errors with additional debugging information.

Cloudflare, “Cloudflare outage on November 18, 2025”, impact table. All times UTC.

The same bad file produced errors on the new proxy and silent false positives on the old one

Unrelated to this incident, customer traffic was being migrated to a new proxy version, FL2.
Both versions were affected by the bad feature file, but the impact each one produced differed.

FL2 · THE NEW PROXY ENGINE

HTTP 5xx errors

Customers deployed on the new FL2 proxy engine observed HTTP 5xx errors for any traffic that depended on the bots module.

This also affected Workers KV and Access, which rely on the core proxy.

FL · THE PREVIOUS PROXY ENGINE

Silent bot-score failure

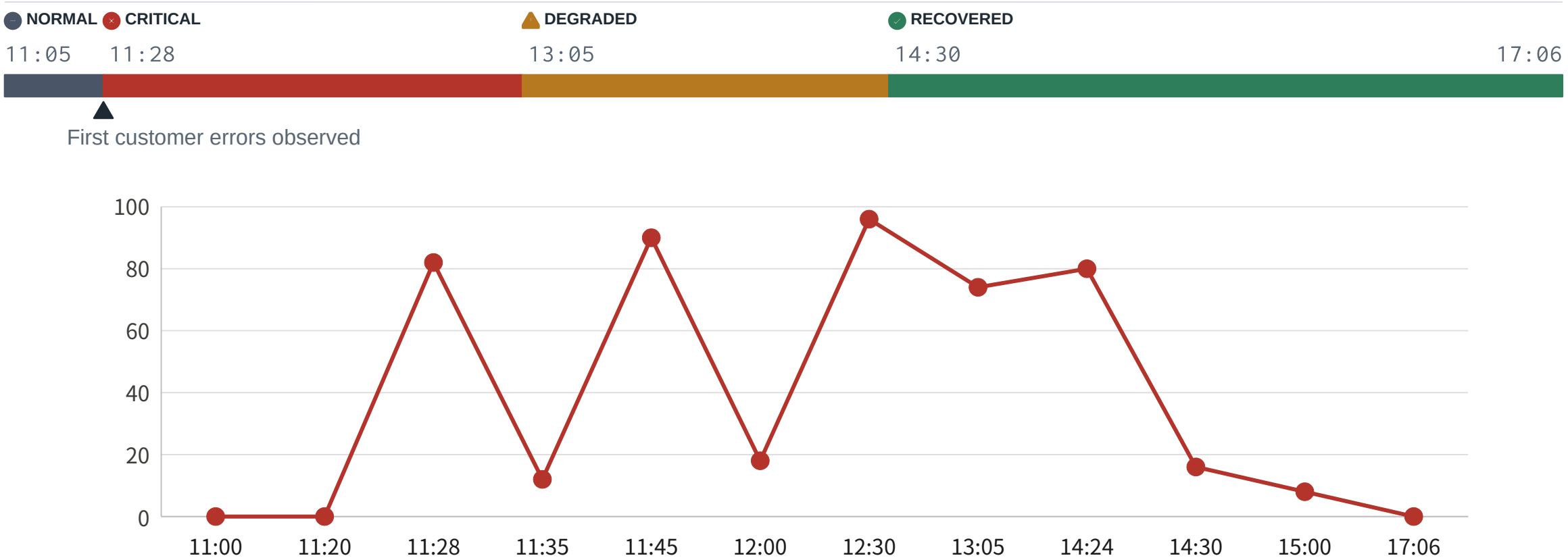
Customers on the old FL engine did not see errors, but bot scores were not generated correctly and all traffic received a bot score of zero.

Customers with rules that block bots would have seen large numbers of false positives.

Customers who were not using the bot score in their rules did not see any impact.

Observed fact: both proxy behaviours are stated in the published account; neither is inferred here.

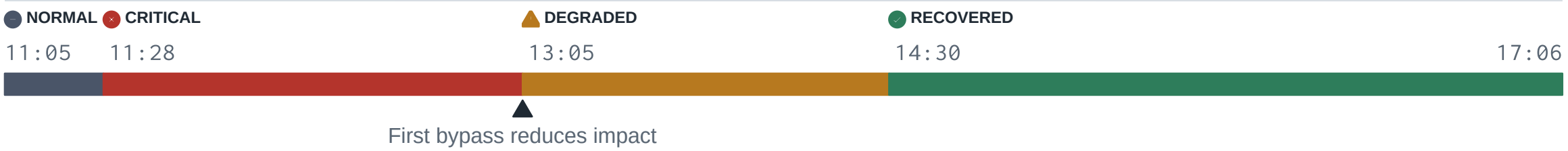
The error rate rose and fell every five minutes until every database node produced the bad file



The feature file was regenerated every five minutes by a query on a ClickHouse cluster that was being gradually updated, so each cycle could produce either a good or a bad file — until every ClickHouse node generated the bad file and the failure stabilised.
Schematic reconstruction from the published account, not the original monitoring record. No error-rate magnitude is published.

Cloudflare, “Cloudflare outage on November 18, 2025”. Severity bands are this review’s reading of the published status column.

From the 11:05 access-control change to the first bypass at 13:05



Time (UTC)	Severity	Recorded event
	Normal	Database access control change deployed.
	Critical	Deployment reaches customer environments; first errors observed on customer HTTP traffic.
	Critical	
	Critical	
	Degraded	

1 h 37 min from the first customer errors at 11:28 to the first bypass at 13:05 — derived from the published timestamps.

Cloudflare, “Cloudflare outage on November 18, 2025”, incident timeline. Severity bands are this review’s reading of the published status column.

From confidence in the trigger at 13:37 to full restoration at 17:06



Time (UTC)	Severity	Recorded event
	Degraded	
	Degraded	
	Degraded	
	Recovered	
	Recovered	All services resolved. All downstream services restarted and all operations fully restored.

3 h 02 min from impact start to main resolution, and 5 h 38 min to full restoration — derived from the published timestamps.

Cloudflare, “Cloudflare outage on November 18, 2025”, incident timeline. Severity bands are this review’s reading of the published status column.

Detection took three minutes; identifying the failing module took two hours and fifty-three

AUTOMATED DETECTION

3 min

11:28 impact starts to 11:31, the first automated test that detects the issue.

TIME TO IDENTIFY THE FAILING MODULE

2 h 53 min

11:31 detection to 14:24, when Bot Management was identified as the source of the 500 errors.

THE FAILURE ALTERNATED

The system failed and then recovered as good and bad files alternated every five minutes — behaviour the published account calls very unusual for an internal error.

THE STATUS PAGE WENT DOWN TOO

Cloudflare's status page, hosted completely off Cloudflare's infrastructure with no dependencies on it, went down at the same time. The published account calls this a coincidence.

RECENT DDoS ACTIVITY

In the internal incident chat room the team was concerned this might continue the recent spate of high-volume Aisuru DDoS attacks.

Inferred sequence: the attack hypothesis is what the published account says the team held at the time. Naming it here is not a criticism of holding it, and the account is explicit that no cyber attack or malicious activity was involved.

Cloudflare, "Cloudflare outage on November 18, 2025", incident timeline and status-page section. All times UTC.

A query that filtered on table name but not database name became an unhandled panic in the core proxy

Access change made explicit → Query returns duplicate rows → Feature file doubles in size → 200-feature limit exceeded → Thread panics, 5xx returned

THE QUERY

```
SELECT
name,
type
FROM system.columns
WHERE table = 'http_requests_features'
order by name;
```

The feature-file query filtered on the table name but not the database name, so after the 11:05 change made access to the r0 database explicit it returned columns twice.

THE LIMIT

```
thread fl2_worker_thread panicked:
called Result::unwrap() on an Err value
```

The bots module preallocates memory for at most 200 features, well above the roughly 60 in use. The oversized file hit that limit and the check produced an unhandled error.

Observed facts: the access change, the query form, the file growth and the 200-feature limit are each stated in the published account. The order that connects them into one chain is that account's own inferred sequence.

Cloudflare, "Cloudflare outage on November 18, 2025", query-behaviour-change and memory-preallocation sections.

Four conditions had to hold at once, and no single one of them is the cause

TECHNICAL

An assumption about the query’s scope was never restated

The feature-generation query assumed table metadata would only ever be returned for the default database. That assumption was not revisited when the permissions model changed at 11:05.

TECHNICAL

A fixed limit guarded memory but had no bounded failure path

The bots module preallocates memory for at most 200 machine-learning features. An input above that limit produced an unhandled error and a thread panic, not a rejected file or a degraded score.

PROCEDURAL

Cloudflare-generated configuration was trusted more than user input

The first published remediation item is to harden ingestion of Cloudflare-generated configuration files in the same way as user-generated input — which states that, before this incident, it was not.

PROCEDURAL

The change propagated gradually, so the failure looked like an attack

Only the updated part of the ClickHouse cluster produced bad data, so output alternated every five minutes and the failure signature resembled an attack rather than a deployment.

Each factor traces to the timeline evidence on slides 6 and 7, and no single one of them is described here as the cause.

Cloudflare, “Cloudflare outage on November 18, 2025”, query-behaviour, memory-preallocation and remediation sections.

Rapid global propagation of internally generated configuration is the condition both 2019 and 2025 share

SYSTEMIC CONDITION

Internally generated configuration reaches every machine in minutes

Both outages began inside Cloudflare’s own configuration path, and in both the step that made a local defect global was the same. Naming that shared condition is an inference across two published accounts.

2 July 2019

TRIGGER

A new WAF managed rule containing a regular expression that backtracked enormously and exhausted CPU worldwide.

PROPAGATION

Pushed through Quicksilver, which distributes changes globally in seconds; by design the WAF skipped the staged rollout process.

MITIGATION

A global kill switch — a “global terminate” — executed at 14:07; traffic and CPU normal again by 14:09.

TIME TO RECOVERY

`27 minutes`

18 November 2025

TRIGGER

A database permissions change that made the Bot Management feature file more than double in size.

PROPAGATION

The feature file is refreshed every few minutes and published to the entire network so the model can react to new bots and new bot attacks.

MITIGATION

No per-feature kill switch is described; the remediation commits to enabling more global kill switches.

TIME TO RECOVERY

`5 h 38 min`

The published account calls this Cloudflare’s worst outage since 2019, and the first in six-plus years to stop most core traffic.

Cloudflare, “Cloudflare outage on November 18, 2025”, and “Details of the Cloudflare outage on July 2, 2019”. All times UTC.

Three controls limited the damage and are the ones to protect

PROTECT THIS

Automated detection was fast

The first automated test detected the issue at 11:31, three minutes after impact began at 11:28. Manual investigation started the following minute and the incident call was created at 11:35.

PROTECT THIS

A bypass existed and was used

At 13:05, internal system bypasses failed Workers KV and Cloudflare Access back to a prior version of the core proxy, reducing impact for every downstream system — before the trigger was known.

PROTECT THIS

The older engine degraded, not failed

Customers on the old FL proxy engine did not see errors, and customers who were not using the bot score in their rules saw no impact at all.

Observed facts from the published account; the judgement that these three are worth protecting is this review’s.

Cloudflare, “Cloudflare outage on November 18, 2025”, incident timeline and other-impact section. All times UTC.

Four corrective actions are published, and none of them carries an owner, a date, or a verification method

Published corrective action	Class	Factor addressed	Owner	Due	Verification
	Prevention		NO DATA	NO DATA	NO DATA
Enabling more global kill switches for features	Mitigation		NO DATA	NO DATA	NO DATA
	Mitigation		NO DATA	NO DATA	NO DATA
	Prevention		NO DATA	NO DATA	NO DATA

The published account lists these four remediation items and no more. It states no owner, no target date and no verification method for any of them, so those three columns are recorded as NO DATA rather than dropped.

Cloudflare, “Cloudflare outage on November 18, 2025”, remediation and follow-up steps. Factor mapping is this review’s.

Five questions the published account does not settle

01 **When did impact begin — 11:20 or 11:28?**

The opening paragraph states the network began experiencing significant failures at 11:20; the incident timeline records impact starts at 11:28. This deck uses 11:28 for every derived duration and says so on each page that shows one.

02 **When was the Workers KV bypass effective — 13:04, 13:05 or 13:10?**

The narrative gives 13:04 for the patch and 13:10 for restoration, while the timeline gives 13:05 for the bypass. Three values describe one mitigation, and the published account does not reconcile them.

03 **What did the outage cost?**

No error count, request volume, affected-customer count, availability figure or financial figure appears anywhere in the published account. NO DATA.

04 **Who owns the four corrective actions, and by when?**

No owner, target date or verification method is published for any of the four remediation items. NO DATA.

05 **Do other core proxy modules share the unbounded failure pattern?**

The account commits to reviewing failure modes for error conditions across all core proxy modules, but does not say whether any others were found. Unconfirmed.

Each item names the evidence that would settle it. None is resolved here by inference. Cloudflare, 18 November 2025 account.

Every figure in this deck traces to one of three published records

PRIMARY SOURCE

Cloudflare outage on November 18, 2025 — Matthew Prince, 18 November 2025

Supports slides 1–10 and 12–14: the timeline, the impact table, the mechanism, the factors, the controls and the actions.

COMPARISON SOURCE

Details of the Cloudflare outage on July 2, 2019

Supports slide 11: the 2019 trigger, the Quicksilver propagation path, the global kill switch and the 27-minute recovery.

OPERATOR STATUS RECORD

Cloudflare System Status

Listed as the operator’s own event record; not consulted for any figure in this deck.

All times in this deck are UTC to the minute, exactly as published. Elapsed durations are arithmetic on published timestamps and are labelled as derived. No figure originates outside these documents; where the account states no value, the page says NO DATA or marks the item unconfirmed. The schematic chart on slide 5 carries no published magnitudes, and the severity bands on slides 5 to 7 are this review’s reading of the published status column.